POAMS: LECTURE 15
POAMS AND THE TWO-BODY PROBLEM

Recall that POAMS seeks to explain the phenomenon of natural orbital motion in purely in terms of angular momentum. As a consequence, it is our Normal Realist replacement of Newton's theory of gravitation, as well as other theories. In this lecture, I shall apply the fundamental postulates of POAMS to the two-body problem, derive the POAMS general equation of motion, and compare this approach to Newtonian theory. This lecture is based on some of Chapter 6 of 'Light-speed....'.

1. The fundamental postulates of the Pope-Osborne Angular Momentum Synthesis (POAMS) [*Recap*]

Recall that the fundamental postulates of the *Pope-Osborne Angular Momentum Synthesis,* known by the acronym POAMS, are as follows:

Postulate 1: *Natural (inertial) motion is always orbital, the natural path of any particle being the trajectory followed by the particle when it moves freely under the influence of nothing but its own angular momentum (angular inertia).*

Postulate 2: *Angular momentum (angular inertia) is holistically conserved.*

Our approach to orbital motion takes Newton's 'force of gravity' out of metaphysical inscrutability and puts it where it truly belongs – that is, in actual physical reality – as the *felt* and *measurable* force required to prevent bodies from travelling in their other naturally force-free (angular inertial) orbits. The POAMS premise is that it is angular momentum which keeps bodies apart, and if angular momentum vanishes, then bodies automatically fall together, in what orthodox physics describes as 'gravitational attraction'.

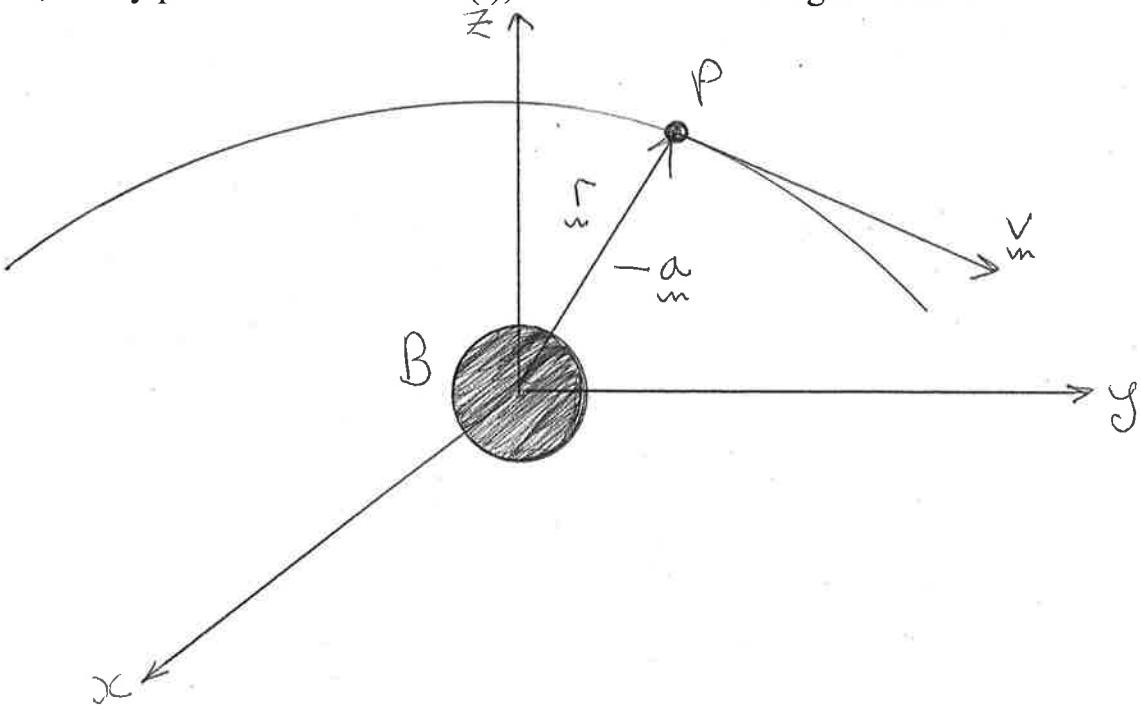
In POAMS, a *constrained* ('unnatural') trajectory represents a state of motion in which a body is held, by a *real measurable* force, against its natural tendency to orbit freely. In our alternative interpretation, the true (*i.e.* measurable) force is not *concealed* in the natural orbit of the body, but is *revealed* in the difference between that natural orbit and a constrained

one. For example, in POAMS, the weight with which a body presses on the surface of Earth is not the result of some unseen and mysterious 'pull of gravity'. It is because the body, where it stands on the rotating Earth, has insufficient angular momentum to orbit freely (weightlessly), so that with the angular momentum it has, the *real* force it exerts on a weighing scale is that of the reaction of Earth's surface preventing it from orbiting where it should, much closer to Earth's centre. [*End of recap*]

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2. <u>The POAMS approach to the two-body problem</u>

In this lecture, we begin to apply POAMS to the study of the so-called *two-body problem*. Recall that this consists of a particle, P, of (non-zero) mass m orbiting a body, B, of mass M, with m very much less than M. (More generally, if m is a significant fraction of M, *then both P and B orbit their combined centre-of-mass.*) All bodies in motion can be considered as part of such a paired system. As in our treatment of the Newtonian approach given in Lecture 13, we let the centre-of-mass of B lie at the origin, O, of a set of Cartesian coordinates and let the position vector of P, in relation to O, at any particular time t be $\mathbf{r}(t)$, as shown in the diagram below.



The POAMS approach to this two-body problem is to use a combination of conservation of angular momentum together with empirical evidence to derive the equation of motion obtained by Newtonian theory, without any

need for an *in vacuo* 'gravitational force'. For the time being, we shall assume that P and B either do not spin, or that they spin very slowly, so that there is negligible contribution from the spin angular momenta of P and B. The full incorporation of spin will be discussed in later lectures. We shall also assume that the system is *isolated*, *i.e.*, not subjected to any external effects. It then follows by Postulate 2 that the angular momentum of P relative to O is always constant.

Recall that the *angular momentum*, $\mathbf{L}$, of the particle P relative to the chosen origin, O, is defined as the vector

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} \qquad (1)$$

[*Lecture 13, equation (2)*] where $\mathbf{v}(t) = \mathrm{d}\mathbf{r}/\mathrm{d}t$ is the velocity of P, which always lies tangentially to the path of P, as indicated in the figure above. Since the vector $\mathbf{L}$ lies perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$, and is constant in time, it follows that the orbit of P *lies in a plane* and that $L = \|\mathbf{L}\|$ is a constant. Also, since $\mathrm{d}\mathbf{L}/\mathrm{d}t = \mathbf{0}$, it follows from definition (1) that

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \times m\mathbf{v} + \mathbf{r} \times m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times m\mathbf{a} = \mathbf{0} \Rightarrow \mathbf{r} \times m\mathbf{a} = \mathbf{0}$$

where $\mathbf{a}(t)$ is P's acceleration relative to O, since $\mathbf{v} \times \mathbf{v} = \mathbf{0}$. Note that $\mathbf{v}(t)$ cannot be a constant in orbital motion, since otherwise the path of P would be rectilinear, and hence $\mathbf{a}(t) \neq \mathbf{0}$. Also, $\mathbf{r}(t) \neq 0$, since otherwise P would be coincident with B. It then follows by the result above that $\mathbf{a}(t)$ must be parallel to $\mathbf{r}(t)$ and so, clearly, the acceleration of P is directed towards O. Compare this approach with the Newtonian treatment given in Section 3 of Lecture 13.

Since the orbit of the particle P lies in a plane, as in Section 3 of Lecture 13, we may employ *plane polar coordinates*, r and θ, where $r = \|\mathbf{r}\|$ and θ is the angle between the radial vector $\mathbf{r}$ and some fixed radial axis. Recall that the *angular speed* of P is $\omega(t) = \mathrm{d}\theta/\mathrm{d}t$. *Exactly* as in Section 3 of Lecture 13, the (orbital) speed, $v(t) = \|\mathbf{v}(t)\|$, of P relative to O is then given by

$$v^2 = \left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + r^2\omega^2 \qquad (2)$$

[*Lecture 13, equation (5)*] and the magnitude of P's angular momentum is

$$L = mr^2\omega \quad (3)$$

[*Lecture 13, equation (7)*] which is constant. The acceleration of P is given by

$$\mathbf{a}(t) = \left(\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - r\omega^2\right)\mathbf{n} \quad (4)$$

where **n** is a vector of unit length in the direction of **r**. [*Lecture 13, equation (6)*]

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3. The POAMS equation of motion

In general, in order to determine all possible orbits of P about B, we require, of course, an *equation of motion*. Since the acceleration of P is directed towards B, this must take the form

$$\mathbf{a}(t) = -h(r,\theta)\mathbf{n} \quad (5)$$

for some function $h(r, \theta)$, where again, **n** is a vector of unit length in the direction of **r.** Equation (4) then provides our general equation of motion. However, there are two main reasons why we discount the case in which h is an explicit function of θ. Firstly, the angle θ is, in a sense, arbitrary on P's trajectory, since it is measured from any chosen radial line. Hence, it does not make physical sense for the acceleration, $\mathbf{a}(t)$, of P to depend on θ. Secondly, we wish equation (4) to provide *circular orbits* as a special case. For the *circle* of radius k, *i.e.*, the circle with equation $r = k$, to be a solution of equation (4), it follows by our hypothesis (5) and equation (3) that

$$h(r,\theta) = r\omega^2 = \frac{L^2}{m^2k^3}$$

so that h cannot depend explicitly on θ.

It then follows from equations (3) and (4) that, in POAMS, the most general equation of motion of P about B is

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - \frac{L^2}{m^2 r^3} = -h(r) \quad (6)$$

for some function, $h(r)$. This, in general, is a second order differential equation for $r(t)$, in which L and m are constants. Equation (6), of course, provides the corresponding Newtonian equation of motion,

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - \frac{L^2}{m^2 r^3} = -\frac{GM}{r^2} \quad (7)$$

[*Lecture 13, equation (9)*] as a special case, where M is the mass of the body B and G is Newton's gravitational constant.

It is easily checked, by differentiating the following equation, that the first integral of equation (6) is

$$\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \frac{L^2}{m^2 r^2} = 2H(r) + C \quad (8)$$

where $\mathrm{d}H/\mathrm{d}r = -h(r)$ and C is a constant [See endnote (*) for details].

Note that from equations (2) and (3), the kinetic energy, K, of P in its orbit is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \frac{L^2}{m^2 r^2}\right)$$

so that from equation (8),

$$K = mH(r) + \frac{Cm}{2}$$

It then follows by conservation of energy (*see Lecture 9, equation (15)*), that the potential energy of P's orbit is $-mH(r)$ and the total energy $E = Cm/2$ is a constant.

For the moment, we shall ignore the special case of 'radial free-fall' for which, as we have seen, $L = 0$ and θ is a constant, and the case of circular orbits, for which r is a constant. Then multiplying equation (8) by $(1/\omega)^2 = (dt/d\theta)^2$, using the chain rule, equation (3) and the expression given above for the total energy, E, of the orbit gives,

$$\left(\frac{dr}{dt}\right)^2\left(\frac{dt}{d\theta}\right)^2 + \frac{L^2}{m^2r^2}\left(\frac{dt}{d\theta}\right)^2 = 2H(r)\left(\frac{dt}{d\theta}\right)^2 + \frac{2E}{m}\left(\frac{dt}{d\theta}\right)^2$$

$$\Rightarrow \left(\frac{dr}{d\theta}\right)^2 + \frac{L^2}{m^2r^2}\left(\frac{m^2r^4}{L^2}\right) = 2H(r)\left(\frac{m^2r^4}{L^2}\right) + \frac{2E}{m}\left(\frac{m^2r^4}{L^2}\right)$$

$$\Rightarrow \left(\frac{dr}{d\theta}\right)^2 + r^2 = \frac{2m^2}{L^2}H(r)r^4 + \frac{2Em}{L^2}r^4 \qquad (9)$$

Finally, it is more convenient to work in terms of the variable $u(t)$, where $u = 1/r$, so that $dr/d\theta = (-1/u^2)du/d\theta$. Making this substitution in equation (9) gives

$$\frac{1}{u^4}\left(\frac{du}{d\theta}\right)^2 + \frac{1}{u^2} = \frac{2m^2}{L^2}\frac{1}{u^4}H(u) + \frac{2Em}{L^2}\frac{1}{u^4}$$

Then letting $\beta = L/m$ and $k = 2Em/L^2$, this equation reduces to,

$$\left(\frac{du}{d\theta}\right)^2 + u^2 = \frac{2}{\beta^2}H(u) + k \qquad (10)$$

In theory, given $H(u)$, this first order differential equation may be solved to provide the equation of P's trajectory relative to B, in the form $u = u(\theta)$.

In general, POAMS does not require that all bounded orbits of P given by the equation of motion (10) are ellipses, as in Newtonian theory. In fact, it will be shown in later lectures, that when time-dilation effects have been taken into account, possible bounded orbits of P about B are not simple ellipses. However, for now, we wish to compare POAMS with Newtonian theory and so require that equation (10) provides *closed* orbits which are *ellipses*, so as to agree with the approximate empirical evidence, as dictated by our Normal Realist approach.

Note that, in terms of Newtonian language, any choice of $h(r)$ in our POAMS equation of motion (6), provides a *central force*. Notice also that equation (6) produces *circular orbits*, given by constant r, for any choice of $h(r)$. The key in determining the specific choice of $h(r)$ here lies in *Bertrand's Theorem* first formulated and proved by Joseph Bertrand (1822-1900) in 1873. This is a theorem from classical mechanics which essentially states that there are only *two* choices for $h(r)$ in (6) for which all bounded orbits given by (10) are also stable *closed* orbits. These are $h(r) = \alpha r$ or $h(r) = \alpha/r^2$, for some constant α. The proof of this result involves classical perturbation techniques using Taylor and Fourier series expansions. Classically speaking, $h(r) = \alpha/r^2$ corresponds to an inverse-square central force, as in Newton's gravitational theory, and $h(r) = \alpha r$ corresponds to the case of a *radial harmonic oscillator.*

We first of all consider the case in which $h(r) = \alpha r$ (α a constant) in equation (6). Then $H(u) = -\alpha/2u^2$ in equation (10) and the general solution to equation (10) in this case can be verified to be

$$u^2(\theta) = \frac{1}{r^2(\theta)} = A \sin 2(\theta - \theta_0) + \frac{k}{2}$$

where θ_0 is a constant and $A^2 = k^2/4 - \alpha/\beta^2$. [See endnote (**) for details] For certain values of the constants, this solution represents a closed curve that it is symmetrical about the radial axis but is not an ellipse. Note that in this case, since $h(r) = \alpha r$, it follows from equation (5) that $\|\mathbf{a}(r)\| = \alpha r$, where $\mathbf{a}$ is the acceleration of P. Hence, $\|\mathbf{a}(r)\|$ takes its *maximum* value at the point on the orbit of P *farthest* from O, which does not agree with the empirical evidence. Therefore, this case may be eliminated.

The only case that remains is then where $h(r) = \alpha/r^2$ in equation (6). In this case, $H(u) = \alpha u$ in equation (10) and its general solution is then

$$u(\theta) = \frac{1}{r(\theta)} = \frac{\alpha}{\beta^2}(1 + e\cos(\theta - \theta_0))$$

where θ_0 and e are constants, with $e^2 = 1 + k\beta^4/\alpha^2$. [See endnote (***) for details]. This is the equation of a *conic section* and represents an *ellipse* if the *eccentricity* $e < 1$, a parabola if $e = 1$ and a hyperbola if $e > 1$. A closed orbit of P around B is therefore possible in this case in the form of an ellipse, when $e < 0$. The special case of a circle is obtained when $e = 0$.

In summary, then, if closed orbits of P about B are required, then the only choice for $h(r)$ in (6) that agrees with the empirical evidence is $h(r) = \alpha/r^2$. Consequently, the POAMS second-order equation of motion for P in a closed orbit about B is,

$$\frac{d^2r}{dt^2} - \frac{L^2}{m^2r^3} = -\frac{\alpha}{r^2} \qquad (11)$$

which, as in the Newtonian case, is to be used in conjunction with

$$\frac{d\theta}{dt} = \frac{L}{mr^2}$$

(see equation (3)). This includes the special cases of 'radial free-fall' when $L = 0$, so that θ is a constant, and *circular* motion, for which r is a constant. Mathematically speaking, the POAMS equation (11) is the same as the Newtonian equation (7), with $\alpha = GM$. Then all the standard results of Newtonian theory follow. However, it is important to emphasise here that equation (11) has now been derived with no reference to arguments involving an *in vacuo* 'force of gravitational attraction', in radical contrast to Newtonian theory.

In the next lecture, I will go on to study special cases of the POAMS equation of motion for the two-body problem, and will provide two specific numerical examples.

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(*) $$\left(\frac{dr}{dt}\right)^2 + \frac{L^2}{m^2 r^2} = 2H(r) + C$$

$$\Rightarrow 2\left(\frac{dr}{dt}\right)\left(\frac{d^2r}{dt^2}\right) - \frac{2L^2}{m^2 r^3}\left(\frac{dr}{dt}\right) = 2H'(r)\left(\frac{dr}{dt}\right)$$

$$\Rightarrow \frac{d^2r}{dt^2} - \frac{L^2}{m^2 r^3} = H'(r) = -h(r)$$

(**) In this case, $H'(r) = -h(r) = -\alpha r$

$$\Rightarrow H(r) = -\frac{\alpha r^2}{2} = \frac{-\alpha}{2u^2} \quad \text{w.l.o.g}$$

Then eqn. (10) reads,

$$\left(\frac{du}{d\theta}\right)^2 + u^2 = -\frac{2\alpha}{2\beta^2 u^2} + k$$

i.e. $$u^2\left(\frac{du}{d\theta}\right)^2 + u^4 = ku^2 - \frac{\alpha}{\beta^2}$$

Now $u^2(\theta) = A\sin 2(\theta - \theta_0) + k/2$

$$\Rightarrow 2u\frac{du}{d\theta} = 2A\cos 2(\theta - \theta_0)$$

$$\Rightarrow u^2\left(\frac{du}{d\theta}\right)^2 + u^4 = A^2\cos^2 2(\theta-\theta_0) + A^2\sin^2 2(\theta-\theta_0) + kA\sin 2(\theta-\theta_0) + k^2/4$$

$$\Rightarrow u^2\left(\frac{du}{d\theta}\right)^2 + u^4 = A^2 + k^2/4 + k\left(u^2 - k/2\right) = A^2 - k^2/4 + ku^2$$

Then $A^2 = k^2/4 - \alpha/\beta^2$

$$\Rightarrow u^2\left(\frac{du}{d\theta}\right)^2 + u^4 = ku^2 - \alpha/\beta^2 \quad \checkmark$$

(***) $h(r) = \frac{\alpha}{r^2} \Rightarrow -H(r) = -\frac{\alpha}{r} = -\alpha u$ w.l.o.g.

Then eqn. (10) reads;

$$\left(\frac{du}{d\theta}\right)^2 + u^2 = \frac{2\alpha}{\beta^2} u + k$$

Now $u(\theta) = \frac{\alpha}{\beta^2}\left(1 + e\cos(\theta - \theta_0)\right)$

$$\Rightarrow \frac{du}{d\theta} = -\frac{\alpha e}{\beta^2}\sin(\theta - \theta_0)$$

$$\therefore \left(\frac{du}{d\theta}\right)^2 + u^2 = \frac{\alpha^2}{\beta^4} e^2 \sin^2(\theta - \theta_0)$$

$$+ \frac{\alpha^2}{\beta^4}\left(1 + 2e\cos(\theta - \theta_0) + e^2\cos^2(\theta - \theta_0)\right)$$

$$= \frac{\alpha^2}{\beta^4}\left(e^2 + 1 + 2\left(\frac{\beta^2}{\alpha} u - 1\right)\right)$$

$$= \frac{2\alpha}{\beta^2} u + \frac{\alpha^2}{\beta^4}\left(e^2 - 1\right)$$

Then $e^2 = 1 + \frac{k\beta^4}{\alpha^2}$ gives

$$\left(\frac{du}{d\theta}\right)^2 + u^2 = \frac{2\alpha}{\beta^2} u + k \quad \checkmark$$